

Questions with Answer Keys

#MathBoleTohMathonGo

Q1 - Dot and Cross Product

Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be vectors of lengths 3, 4, 5 respectively and $\mathbf{a}$ be perpendicular to $(\mathbf{b} + \mathbf{c})$, $\mathbf{b}$ to $(\mathbf{c} + \mathbf{a})$ and $\mathbf{c}$ to $(\mathbf{a} + \mathbf{b})$, then the value of $(\mathbf{a} + \mathbf{b} + \mathbf{c})$ is

(1) $2\sqrt{5}$

(2) $2\sqrt{2}$

(3) $10\sqrt{5}$

(4) $5\sqrt{2}$

Q2 - Dot and Cross Product

Let $\mathbf{b} = 4\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ and $\mathbf{c}$ be a vector perpendicular to $\mathbf{b}$ and it lies in the XY -plane. A vector in the XY -plane having projections 1 and 2 along $\mathbf{b}$ and $\mathbf{c}$ is

(1) $2\hat{\mathbf{i}} + \hat{\mathbf{j}}$

(2) $\hat{\mathbf{i}} - 2\hat{\mathbf{j}}$

(3) $\frac{1}{5}(-2\hat{\mathbf{i}} + 11\hat{\mathbf{j}})$

(4) None of these

Q3 - Dot and Cross Product

Two adjacent sides of a parallelogram $ABCD$ are given by $\mathbf{AB} = 2\hat{\mathbf{i}} + 10\hat{\mathbf{j}} + 11\hat{\mathbf{k}}$ and

$\mathbf{AD} = -\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$. The side AD is rotated by an acute angle α in the plane of parallelogram so that AD becomes AD' . If AD' makes a right angle with the side AB , then the cosine of angle α is

(1) $\frac{8}{9}$

(2) $\frac{\sqrt{17}}{9}$

(3) $\frac{1}{9}$

(4) $\frac{4\sqrt{5}}{9}$

Q4 - Dot and Cross Product

If vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ represent sides BC , CA and AB of $\triangle ABC$, then

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(1) $\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} = 0$

(2) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a}$

(3) $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{a}$

(4) $\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} = 0$

Q5 - Dot and Cross Product

Let $\mathbf{OA} = \mathbf{a}$, $\mathbf{OB} = 10\mathbf{a} + 2\mathbf{b}$ and $\mathbf{OC} = \mathbf{b}$, where O , A and C are non-collinear points, p denotes the area of quadrilateral $OABC$ and q denotes the area of the parallelogram with OA and OC as adjacent sides. If $p = kq$, then k is equal to

(1) 5

(2) 6

(3) 4

(4) 3

Q6 - Dot and Cross Product

Let $\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\mathbf{b} = 2\hat{\mathbf{i}} - \hat{\mathbf{k}}$. The point of intersection of the lines $\mathbf{r} \times \mathbf{a} = \mathbf{b} \times \mathbf{a}$ and $\mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$ is

(1) $-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$

(2) $3\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}$

(3) $3\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$

(4) $\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$

Q7 - Dot and Cross Product

Unit vector perpendicular to the plane of $\triangle ABC$ with position vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ of the vertices A , B , C is

(1) $\frac{\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}}{\Delta}$

(2) $\frac{\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}}{2\Delta}$

(3) $\frac{\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}}{4\Delta}$

(4) None of these

Questions with Answer Keys

#MathBoleTohMathonGo

Q8 - Scalar Triple Product and Vector Triple Product

The volume of parallelopiped whose sides are given by $\mathbf{OA} = 2\hat{i} - 2\hat{j}$, $\mathbf{OB} = \hat{i} + \hat{j} - \hat{k}$, $\mathbf{OC} = 3\hat{i} - \hat{k}$ is

- (1) 2
- (2) 3
- (3) $\frac{2}{7}$
- (4) $\frac{4}{13}$

Q9 - Scalar Triple Product and Vector Triple Product

For three vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, the value of $[\mathbf{a} + \mathbf{b}, \mathbf{b} + \mathbf{c}, \mathbf{c} + \mathbf{a}]$ is

- (1) $[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$
- (2) 0
- (3) $[\mathbf{abc}]^2$
- (4) $2[\mathbf{abc}]$

Q10 - Scalar Triple Product and Vector Triple Product

Let $\mathbf{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\mathbf{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\mathbf{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $\mathbf{c}$ is a unit vector perpendicular to both the vectors $\mathbf{a}$ and $\mathbf{b}$. If the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{6}$, then

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2 \text{ is}$$

- (1) 0
- (2) $\frac{1}{4}(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$
- (3) 1
- (4) $\frac{3}{4}(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)(c_1^2 + c_2^2 + c_3^2)$

Q11 - Scalar Triple Product and Vector Triple Product

Let $\mathbf{a} = \hat{i} + \hat{j} + \hat{k}$, $\mathbf{b} = \hat{i} - \hat{j} + \hat{k}$ and $\mathbf{c} = \hat{i} - \hat{j} - \hat{k}$ be three vectors. A vector $\mathbf{v}$ in the plane of $\mathbf{a}$ and $\mathbf{b}$ whose projection on $\mathbf{c}$ is $\frac{1}{\sqrt{3}}$, is given by

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(1) $\hat{i} - 3\hat{j} + 3\hat{k}$

(2) $-3\hat{i} - 3\hat{j} - \hat{k}$

(3) $3\hat{i} - \hat{j} + 3\hat{k}$

(4) None of these

Q12 - Scalar Triple Product and Vector Triple Product

Let $\mathbf{a} = \hat{i} - \hat{j}$, $\mathbf{b} = \hat{j} - \hat{k}$ and $\mathbf{c} = \hat{k} - \hat{i}$. If $\mathbf{d}$ is a unit vector such that $\mathbf{a} \cdot \mathbf{d} = 0 = (\mathbf{bcd})$ then $\mathbf{d}$ is equal to

(1) $\pm \frac{(\hat{i} + \hat{j} - 2\hat{k})}{\sqrt{6}}$

(2) $\pm \frac{(\hat{i} + \hat{j} - \hat{k})}{\sqrt{3}}$

(3) $\pm \frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}}$

(4) $\pm \hat{k}$

Q13 - Scalar Triple Product and Vector Triple Product

If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are non-coplanar unit vectors such that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \frac{1}{\sqrt{2}}(\mathbf{b} + \mathbf{c})$, then the angle between $\mathbf{a}$ and $\mathbf{b}$ is

(1) $\frac{3\pi}{4}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{2}$

(4) π

Q14 - Scalar Triple Product and Vector Triple Product

Let $\mathbf{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\mathbf{b} = \hat{i} + \hat{j}$. If $\mathbf{c}$ is a vector such that $\mathbf{a} \cdot \mathbf{c} = |\mathbf{c}| |\mathbf{c} - \mathbf{a}| = 2\sqrt{2}$ and angle between $(\mathbf{a} \times \mathbf{b})$ and $\mathbf{c}$ is 30° , then $|(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}|$ is equal to

(1) $\frac{2}{3}$

(2) $\frac{3}{2}$

(3) 2

(4) 3

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

Q15 - Line

If the straight lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ intersect at a point, then the integer k is

(1) -5

(2) 5

(3) -2

(4) 2

Q16 - Line

If a plane cuts off intercepts $OA = a$, $OB = b$, $OC = c$ from the coordinate axes, then the area (in sq units) of $\triangle ABC$ is

(1) $\frac{1}{2}\sqrt{b^2c^2 + c^2a^2 + a^2b^2}$

(2) $\frac{1}{2}(bc + ca + ab)$

(3) $\frac{1}{2}abc$

(4) $\frac{1}{2}\sqrt{(b-c)^2 + (c-a)^2 + (a-b)^2}$

Q17 - Line

The cartesian form of the equation of a line whose vector form is given by

$$r = (2\hat{i} - \hat{j} + 4\hat{k}) + \lambda(\hat{i} + \hat{j} - 2\hat{k})$$

(1) $\frac{x-2}{1} = \frac{y+1}{1} = \frac{z-4}{-2}$

(2) $\frac{x+2}{1} = \frac{y-1}{1} = \frac{z+4}{-2}$

(3) $\frac{x-2}{-1} = \frac{y+1}{1} = \frac{z-4}{-2}$

(4) $\frac{x+2}{1} = \frac{y-1}{-1} = \frac{z+4}{-2}$

Q18 - Line

Let the points $A(a, b, c)$ and $B(a', b', c')$ be at distances r and r' from the origin. The line AB passes through the origin, when

Questions with Answer Keys

#MathBoleTohMathonGo

(1) $\frac{a'}{a} = \frac{b'}{b} = \frac{c'}{c}$

(2) $aa' + bb' + cc' = rr'$

(3) $aa' + bb' + cc' = r^2 + (r')^2$

(4) None of these

Q19 - Line

The vector equations of the two lines L_1 and L_2 are given

$$L_1 : \mathbf{r} = (2\hat{\mathbf{i}} + 9\hat{\mathbf{j}} + 13\hat{\mathbf{k}}) + \lambda(\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}})$$

$$L_2 : \mathbf{r} = (-3\hat{\mathbf{i}} + 7\hat{\mathbf{j}} + p\hat{\mathbf{k}}) + \mu(-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}})$$

then the lines L_1 and L_2 are(1) skew lines for all $p \in R$ (2) intersecting for all $p \in R$ and the point of intersection is $(-1, 3, 4)$ (3) intersecting lines for $p = -2$ (4) intersecting for all real $p \in R$

Q20 - Plane

The reflection of the point $(2, -1, 3)$ in the plane $3x - 2y - z = 9$ is

(1) $\left(\frac{26}{7}, \frac{15}{7}, \frac{17}{7}\right)$

(2) $\left(\frac{26}{7}, -\frac{15}{7}, \frac{17}{7}\right)$

(3) $\left(\frac{15}{7}, \frac{26}{7}, -\frac{17}{7}\right)$

(4) $\left(\frac{26}{7}, \frac{17}{7}, -\frac{15}{7}\right)$

Q21 - Plane

The distance of the point $(-1, -5, -10)$ from the point of intersection of the line, $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 5$, is

(1) 10

(2) 11

Questions with Answer Keys

#MathBoleTohMathonGo

(3) 12

(4) 13

Q22 - Plane

The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$, is

(1) 1

(2) $\frac{7}{6}$ (3) $\frac{6}{7}$

(4) None of these

Q23 - Plane

The acute angle that the vector $2\hat{i} - 2\hat{j} + \hat{k}$ makes with the plane contained by the two vectors $2\hat{i} + 3\hat{j} - \hat{k}$ and $\hat{i} - \hat{j} + 2\hat{k}$ is given by

(1) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (2) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (3) $\tan^{-1}(\sqrt{2})$ (4) $\cot^{-1}(\sqrt{2})$

Q24 - Plane

A point moves so that the sum of the squares of its distance from the six faces of a cube given by

$x = \pm 1, y = \pm 1, z = \pm 1$ is 10 units. The locus of the point is

(1) $x^2 + y^2 + z^2 = 1$ (2) $x^2 + y^2 + z^2 = 2$ (3) $x + y + z = 1$ (4) $x + y + z = 2$

Questions with Answer Keys

#MathBoleTohMathonGo

Q25 - Plane

Two system of rectangular axes have the same origin. If a plane cuts them at distances a, b, c and a_1, b_1, c_1 from the origin, then

$$(1) \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{(a_1)^2} + \frac{1}{(b_1)^2} + \frac{1}{(c_1)^2}$$

$$(2) \frac{1}{a^2} - \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{(a_1)^2} - \frac{1}{(b_1)^2} + \frac{1}{(c_1)^2}$$

$$(3) a^2 + b^2 + c^2 = (a_1)^2 + (b_1)^2 + (c_1)^2$$

$$(4) a^2 - b^2 + c^2 = (a_1)^2 - (b_1)^2 + (c_1)^2$$

Q26 - Plane

If the lines $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$, $\frac{x-1}{3} = \frac{y-2}{-1} = \frac{z-3}{4}$ and $\frac{x+k}{3} = \frac{y-1}{2} = \frac{z-2}{h}$ are concurrent, then

$$(1) h = -2 \text{ and } k = -6$$

$$(2) h = \frac{1}{2} \text{ and } k = 2$$

$$(3) h = 6 \text{ and } k = 2$$

$$(4) h = 2 \text{ and } k = \frac{1}{2}$$

Q27 - Plane

The equation of the plane containing the lines $\mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}$ and $\mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}$ is

$$(1) \mathbf{r} \cdot (\mathbf{a}_1 - \mathbf{a}_2) \times \mathbf{b} = [\mathbf{a}_1 \mathbf{a}_2 \mathbf{a}_3]$$

$$(2) \mathbf{r} \cdot (\mathbf{a}_2 - \mathbf{a}_1) \times \mathbf{b} = [\mathbf{a}_1 \mathbf{a}_2 \mathbf{b}]$$

$$(3) \mathbf{r} \cdot (\mathbf{a}_1 + \mathbf{a}_2) \times \mathbf{b} = [\mathbf{a}_2 \mathbf{a}_1 \mathbf{b}]$$

(4) None of the above

Q28 - Sphere

The radius of circle in which the sphere $x^2 + y^2 + z^2 + 2x - 2y - 4z - 19 = 0$ is cut by the plane

$$x + 2y + 2z + 7 = 0, \text{ is}$$

$$(1) 4$$

Questions with Answer Keys

#MathBoleTohMathonGo

(2) 1

(3) 2

(4) 3

Q29 - Sphere

The shortest distance from the plane $12x + 4y + 3z = 327$ to the sphere $x^2 + y^2 + z^2 + 4x - 2y - 6z = 155$ is

(1) 39

(2) 26

(3) $11\frac{4}{13}$

(4) 13

Q30 - Sphere

The equation of sphere passing through the point $(1, 3, -2)$ and the circle $x^2 + y^2 + z^2 = 25, x = 0$ is

(1) $x^2 + y^2 + z^2 + 11x + 25 = 0$

(2) $x^2 + y^2 + z^2 - 11x + 25 = 0$

(3) $x^2 + y^2 + z^2 + 11x - 25 = 0$

(4) $x^2 + y^2 + z^2 - 11x - 25 = 0$

